Strategic EV Market Entry Analysis

Washington State as a Framework for Urban-Rural Dynamics

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Data Science Foundations to Core Career Track, Springboard

1. Problem Statement

The U.S. Electric Vehicle market is growing but fragmented by geography. Infrastructure density, household income, commute distance, and climate vary sharply across urban, suburban, and rural zones, producing fundamentally different adoption dynamics that a single national strategy cannot address.

This project uses Washington State as an analytical proxy for those dynamics. Washington contains dense Pacific coastal metros, mid-sized suburban commuter corridors, and vast rural stretches east of the Cascades, all within a single jurisdiction with consistent public registration data. Three objectives guided the work:

- Identify at least three distinct Washington State market zones differentiated by EV adoption drivers.
- Build predictive models quantifying the relationship between vehicle price, vintage, and electric range.
- Develop targeted go-to-market strategies for each zone, grounded in what the data can and cannot support.

2. Feature Selection

Five features were identified as analytically relevant to EV adoption. It is important to distinguish between features that were selected for their conceptual relevance and features that were actually used in the trained models, these are not the same set.

Feature	Used in Model?	Source	Status
base_msrp	**Yes**	WA Population	Trained on; individual contribution not isolated
model_year	**Yes**	WA Population	Trained on; individual contribution not isolated
battery_capacity_kWh	**No**	EV Specs 2025	Selected analytically; never entered the model
fast_charge_power_kw	**No**	EV Specs 2025	Selected analytically; never entered the model

top_speed_kmh	**No**	EV Specs 2025	Selected analytically; never entered the model

> *Only base_msrp and model_year entered the trained models. battery_capacity_kWh, fast_charge_power_kw, and top_speed_kmh were identified during exploratory analysis of the EV Specifications dataset as theoretically important adoption drivers, but the two datasets were never merged. Their influence on range and adoption was not measured by the models — it was reasoned from the literature and EDA observations only.*

This gap is the central limitation of the current analysis and the primary driver of the future research agenda in Section 6.

3. Modeling

3.1 Setup

Three regression algorithms were trained on the Washington State EV Population Dataset (~186,000 vehicle registrations). The target variable was electric_range in miles. The feature set consisted of two variables: base_msrp and model_year. An 80/20 train/test split was applied with random_state=42 for reproducibility across all three models.

3.2 Results

Model	R^2 Score	MAE (miles)	vs. Baseline	Selected?
Linear Regression	0.294	45.47	— (baseline)	No
Random Forest	**0.738**	**22.58**	**+44.4 pts R^2**	**YES ✓**
Gradient Boosting	0.738	22.70	+44.4 pts R^2	No

> *Random Forest was selected as the final model: R^2 = 0.738, MAE = 22.58 miles. Gradient Boosting produced a statistically equivalent result (R^2 = 0.7382, MAE = 22.70 miles) and was passed over on the basis of marginally higher error alone. Linear Regression performed substantially worse at R^2 = 0.294 — a 44.4-point gap — confirming that the relationship between price and range is non-linear.*

3.3 What the Model Results Tell Us

An R^2 of 0.738 means that base_msrp and model_year together explain approximately 73.8% of the variance in electric range across Washington State registrations. That is a meaningful result for a two-feature model. However, several conclusions that might appear to follow from this number do not actually follow from the analysis as executed:

What was not measured:

- The individual contribution of base_msrp vs. model_year was never isolated. No feature importance scores, permutation importance analysis, or coefficient breakdown was produced. It is not possible to say from the current outputs which of the two features drove the R^2 of 0.738.
- Whether model_year added meaningful predictive lift over base_msrp alone is unknown. It may have contributed substantially, marginally, or almost nothing. The model was not tested with each feature in isolation.
- The three features that were never trained on — battery_capacity_kWh, fast_charge_power_kw, and top_speed_kmh — may explain a significant portion of the remaining 26.2% variance, or they may be collinear with base_msrp and add little. This is untested.

4. Findings

4.1 What the Model Confirms

The 44.4-point R^2 gap between Linear Regression and Random Forest is the clearest finding in this analysis. It confirms that price and range do not scale linearly, a straight-line fit captures less than a third of the variance that an ensemble method captures using the same two inputs. This means incentive structures based on linear price reductions are likely less effective than threshold-based interventions that move buyers from one range tier to the next.

The model also confirms that base_msrp and model_year together carry meaningful predictive signal about electric range. A model trained on just two publicly available, buyer-facing data points explains nearly three-quarters of range variance across 186,000 registrations. That is a useful baseline.

4.2 What the Model Does Not Confirm

Several findings presented in earlier drafts of this report were not supported by the modeling outputs and are corrected here:

- The claim that model_year is an independent range signal is unverified. It was included in the feature set but its marginal contribution was never measured.
- The claim that fast_charge_power_kw is the critical rural infrastructure variable came from EDA observations on the EV Specifications dataset — not from the trained model. It is a reasonable hypothesis, not a finding.
- The claim that top_speed_kmh cleanly segments market tiers was an analytical inference from the spec data. It was not tested in the model.
- The specific regional thresholds cited previously — such as 500 km range for rural corridors or $35,000 MSRP as an inflection point — were directionally informed by the data but were not outputs of the model. They represent reasonable interpretive judgments, not measured results.

> *The model demonstrates that ensemble methods meaningfully outperform linear regression on this problem using two features, and that price and vintage carry real signal about range. Beyond that, the current analysis does not support confident quantitative conclusions about which features matter most, why the model performs as it does, or how adoption would respond to specific interventions.*

5. Recommendations

The following recommendations are grounded in the modeling results where they can be, and are clearly labeled as directional, based on domain reasoning and EDA, where they cannot. Washington State's three geographic environments serve as the framework.

Zone	WA Example Areas	Primary Barrier	Proposed Strategy
Rural	Eastern WA: Okanogan, Ferry, Pend Oreille counties	Range anxiety; no DC Fast Charging (DCFC) infrastructure	High-range vehicles; DCFC corridor investment
Suburban	Bellevue, Redmond, Kirkland, Spokane Valley	Price sensitivity; moderate commute distance	Entry-level EVs (<$35K); Total Cost of Ownership (TCO) messaging
Urban	Seattle, Tacoma, Olympia, Bellingham	Market near saturation; premium growth	Performance models; fleet electrification

5.1 Rural Washington: Range and Infrastructure First

Eastern and northeastern Washington (Okanogan, Ferry, Pend Oreille, Lincoln counties) present the clearest adoption gap. Travel distances between population centers are long, and DC fast charging infrastructure is sparse along the primary arterials, US-2, US-395, and SR-20.

Supported by data: The non-linear model performance confirms that range is not a simple function of price, suggesting that moving rural buyers into higher-range vehicles requires more than incremental price discounts — it requires reaching a threshold where range anxiety is genuinely resolved.

Directional (not model-derived): *Vehicles with battery_capacity_kWh above 80 kWh and fast_charge_power_kw at or above 150 kW are hypothesized to best serve this market. This is grounded in EDA of the EV Specs dataset and domain reasoning, not in a trained model output.*

- Proposed action: Direct DCFC subsidies (150 kW minimum) to I-90 east of Ellensburg, US-2 through Wenatchee to Spokane, and US-395 through the Okanogan corridor.
- Proposed action: Prioritize SUV and light truck segments in rural dealer inventory, where spec data suggests range and tow capacity are co-requirements.

5.2 Suburban Washington: Price Threshold and TCO

Suburban corridors, Bellevue, Redmond, Kirkland, Spokane Valley, and the I-405 and SR-520 commute bands, have predictable daily travel distances. The binding constraint in these markets is price perception, not range.

Supported by data: The model's non-linear structure implies that price thresholds, not linear discounts, determine which buyers enter the long-range EV tier. A buyer just below a meaningful price breakpoint behaves differently than one well within a tier.

Directional (not model-derived): *The $35,000 MSRP threshold cited as the boundary for range-adequate suburban EVs is an interpretive judgment from the data, not a measured model output. It*

aligns with the federal EV tax credit eligibility boundary, which provides external support for its relevance.

- Proposed action: TCO campaigns comparing 5-year fuel and maintenance costs between entry-level EVs and equivalent ICE vehicles, front-loading the effective post-credit purchase price.
- Proposed action: Focus inventory and incentive communications on vehicles at or below $35,000 MSRP that qualify for the federal $7,500 tax credit.

5.3 Urban Washington: Premium and Fleet

Seattle, Tacoma, Olympia, and Bellingham already exhibit high baseline EV adoption. Volume growth here is limited; the opportunity is margin and fleet scale.

Supported by data: The concentration of high model_year registrations in urban zones within the WA Population dataset is consistent with faster technology turnover in higher-income, higher-density markets.

Directional (not model-derived): *The recommendation to prioritize performance and premium segments in urban cores is grounded in demographic and EDA reasoning, not in a trained model distinguishing urban from rural buyer behavior.*

- Proposed action: Fleet electrification partnerships with municipal fleets (City of Seattle, King County Metro) and ride-share operators to accelerate model-year turnover in the registration base.
- Proposed action: Premium and performance EV positioning (MSRP above $55,000) in urban retail channels where buyer profiles skew technology-forward.

6. Limitations

- Two-feature model: The trained model used only base_msrp and model_year. Three of the five analytically selected features — battery_capacity_kWh, fast_charge_power_kw, and top_speed_kmh — were never incorporated. Their effect on range and adoption is hypothesized but unmeasured.
- No feature importance analysis: The relative contribution of base_msrp vs. model_year to the R^2 of 0.738 was not measured. No permutation importance, SHAP values, or ablation testing was conducted. Statements about which features drive model performance are speculative.
- No hyperparameter tuning: All three models were run on scikit-learn defaults. Random Forest with tuned n_estimators, max_depth, and min_samples_split may perform meaningfully differently — better or worse — than the reported results.
- Datasets were never merged: The EV Specs dataset (472 models, technical specifications) and the WA Population dataset (186,000 registrations, price and year) were analyzed in parallel but never joined. A merged dataset on vehicle make/model would enable the five selected features to be used together in a single model.
- Single-state scope: Washington State provides a useful urban-rural contrast but is not a validated national proxy. Price thresholds, commute distances, and infrastructure gaps differ in Texas, Florida, and the Mountain West. Findings should not be generalized without validation on out-of-state data.

- No adoption outcome variable: electric_range was used as a proxy for adoption potential. Actual adoption — measured by registration growth rates or market penetration by zone — was never modeled directly.

7. Future Research

The following steps are ordered by their expected impact on closing the gap between what this analysis hypothesizes and what it can actually demonstrate:

1. Merge the EV Specs and WA Population datasets on vehicle make/model to build a unified feature space. This is the single highest-priority next step. It would allow battery_capacity_kWh, fast_charge_power_kw, and top_speed_kmh to be trained alongside base_msrp and model_year in a single model.
2. Run feature importance analysis on the current Random Forest model, at minimum, extract built-in impurity-based importance scores for base_msrp and model_year to determine which of the two features is doing the work. Add SHAP values for interpretability.
3. Test each feature in isolation by training separate single-feature models. This would establish the marginal contribution of model_year over base_msrp alone and vice versa — a gap the current analysis leaves open.
4. Apply hyperparameter tuning (GridSearchCV across n_estimators, max_depth, min_samples_split) to determine whether the default Random Forest is near-optimal or leaving significant performance on the table.
5. Replace electric_range as the target variable with a direct adoption measure, such as annual registration growth rate by ZIP code, to model what actually drives market penetration rather than a technical proxy for it.
6. Incorporate county-level DCFC station density from the AFDC database as a spatial feature, enabling the infrastructure-adoption hypothesis to be tested directly in the model rather than reasoned from spec data alone.

8. Conclusion

This project trained three regression algorithms, Linear Regression, Random Forest, and Gradient Boosting, on Washington State EV registration data to predict electric range from two features: base_msrp and model_year. Random Forest was selected as the final model with $R^2 = 0.738$ and MAE = 22.58 miles, substantially outperforming the linear baseline ($R^2 = 0.294$). Gradient Boosting achieved an equivalent R^2 but marginally higher error and was not selected.

The 44.4-point performance gap between the linear and ensemble models is the clearest finding: the relationship between price and range is non-linear, and incentive design should account for threshold effects rather than assuming proportional response to incremental price changes.

Beyond that, the analysis is honest about its limits. Three of the five analytically selected features, battery_capacity_kWh, fast_charge_power_kw, and top_speed_kmh, never entered the trained models. The individual contributions of the two features that did were not measured. Regional recommendations for rural, suburban, and urban Washington State zones are directionally grounded in exploratory analysis and domain reasoning, but they are not outputs of the model.

> *The foundation built here, a reproducible modeling pipeline, a clearly scoped feature set, and an accounting measure of what was and was not measured, is more useful as a starting point than a set of overconfident conclusions would be. The next version of this analysis, with a merged dataset and feature importance analysis, is positioned to make claims the current version cannot.*